

Examining Identity as a Variable of Health Technology Research for Older Adults: A Systematic Review

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ABSTRACT

Innovations in HCI research of health-related pervasive and ubiquitous technologies can potentially improve older adults' access to healthcare resources and support long-term independence in the home. Despite efforts to include their voices in technology research and design, many older adults have yet to actualize these health benefits, with barriers of access and proficiency actually widening the gap of health inequities. We reviewed 174 HCI publications through a systematic review to examine who is engaged in the design of health technologies for older adults, methods used to engage them, and how different types of participation might impact design directions. Findings highlight that thus far, many identity dimensions have not been explored in HCI aging research. We identify research gaps and implications to promote expanding research engagement with these dimensions as a way to support the design of health technologies that see better adoption among marginalized populations.

CCS CONCEPTS

• **Social and professional topics** → **Race and ethnicity; User characteristics**; • **Human-centered computing** → **Interaction design process and methods; HCI design and evaluation methods**.

KEYWORDS

older adults, health, pervasive and ubiquitous technologies, study participants, systematic review, cultural diversity, diversity dimensions

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1 INTRODUCTION

As a research discipline, human-computer interaction (HCI) has increased the number of computing technologies that support older adults' health and wellness, exploring how emerging innovations

such as artificial intelligence, virtual and augmented reality, synchronous tracking, and other ubiquitous systems might meet the needs of the increasing population of seniors. Aging trends in the U.S. suggest that the number of older adults over the age of 65 will reach 83.7 million by 2050 [31]. Globally, older adults make up 8.5% of the overall population, and that statistic will increase exponentially in the next 30 years [32]. Pervasive and ubiquitous computing technologies can play a vital role in older adults' everyday experiences, helping them maintain independent yet socially connected lives while remaining in their homes [112, 131, 186]. Many health technologies now support tracking and self-management of long-term and chronic health conditions, illnesses, and diseases mapping age-related decline among older adults against system capabilities to keep adults healthier for longer. These technologies have also become more personal [51], making critical information to support self-maintenance readily available, gamifying health in attempts to make physical activity more accessible [16, 107], as well as promoting social connections important for individuals living alone [38]. However, as the older adult population increases, so will the need to understand how older adults interact with and accept emerging health-related technologies.

Despite the benefits of health-related technologies for older adults, lower numbers of older adults accept and adopt these technologies compared to the general public [112], and for marginalized and vulnerable populations, acceptance and adoption statistics are often lower. For example, we know certain subgroups of older adults fail to adopt health technologies due to barriers of digital literacy, educational background, and technology accessibility [46, 181]. While these barriers could easily be associated with the larger population of older adults, researchers that employ a social science perspective note that older adults are not a monolith or homogeneous group. Thus, the interaction between older users and technology cannot rely on the premise of age alone, a premise that is often deficit-based. More specifically, barriers to technology adoption are often exacerbated for individuals from historically marginalized or racial minority groups, as well as individuals marginalized along identity dimensions such as gender, social status, economic status, cultural background, sexual identity and orientation, personal and community history, and technological literacy and proficiency [112, 116, 127, 224]. As the computing discipline becomes more concerned with the diversity and inclusion of those historically excluded from research [58, 64, 111, 113], it is critical to understand the dimensions of identity considered in the design of health-related pervasive and ubiquitous technology research for older adults. Current statistics [112] suggesting less participation among certain populations of older adults, beg the question of whether we are focusing on appropriate constructs of aged identities and engaging the widest population of older adults in our research praxis.



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Considering dimensions of identity within HCI research has allowed for a critical reflection on our work's reach in terms of who is engaged [113, 206], and in what ways recent calls for diversity are implemented in our research praxis. Leveraging insights from the social sciences suggests that we must understand people's relation to technology in society and the world's phenomena [185, 186]. Under this presumption, usability considers more than the deficit-based narratives of the challenges of aging, which can often lead to design approaches that emphasize solutionism (i.e., fixing the problem of aging). Instead this perspective urges us to understand dimensions of identity that have an equal impact on the experiences that individuals have with technology, better capturing how a person experiences the world. The adoption of technologies by older adults depends not only on the usability of the design but also on the users' preferences and prior experiences, which are informed by a better understanding of older adults themselves. Yet, one consideration that has often not been explored in health technology research is the examination of dimensions of identity among certain subgroups of the larger aging population and how this may impact technology needs and experiences. The neglect of certain identities in technology design can perpetuate further health inequities, and thus it is critical to advocate for the inclusion of such dimensions in our work. Applying this lens to the study of health-related technologies, we see a need to engage more with the dimensions of identity when researching the development and evaluation of health technologies.

This systematic review aims to understand the constructs examined in previous health technologies research for older adults. We illustrate the current trends and the trajectory of this representation across prominent computing publications. This review does not aim to show a complete picture of all existing literature related to the subject across various sub-fields. Instead, we focus on HCI literature to identify how technology design research is conducted with older adults, specifically, those historically disenfranchised and typically neglected in technology design. By analyzing literature to examine processes and dimensions of identity, we contribute a better understanding of approaches being used to uncover older adults' values, beliefs, and perceptions of health technologies. This review critically reflects on how HCI researchers seek to engage in the design and evaluation of health-related technologies to support older adults' health and well-being by examining the inclusiveness of subgroups within the older adult population. An additional contribution of this systematic review is a methodological assessment of the status quo of aging research in HCI and how this structures design's role in health equity. We look to expand the considerations that are commonly associated with aging-technology research and extend the narrative to include other facets of identity by addressing the following research questions:

- What are the trends in the types of research studies (generative, evaluative) being published about the design of health-related pervasive and ubiquitous technologies for older adults?
- What methods are used in technology design across these studies, and in what ways are subgroups of older adults engaged in the design process?
- Which identity dimensions are reported and examined in health-related HCI research with older adults and in what

ways do HCI researchers understand and categorize older adults as end users? What is the diversity among this population in health-related HCI research?

We discuss recommendations for research practices to ensure HCI researchers are moving towards developing systems for health equity.

2 RELATED WORK

To frame the contribution of this systematic review, we highlight areas of prior work that this review builds upon and expands: 1- trends in health technology research for older adults; 2- existing narratives of older adults in HCI research; and 3- the current push to explore dimensions of identity in HCI.

2.1 Examining Healthcare Technology Trends Among the Aging Population

There have been several efforts to understand healthcare technology trends among the aging population. For example, Vedel and colleagues examined Health Information Technology (HIT) literature in geriatrics and gerontology to identify key focus areas, and their impact on health outcomes [237]. Their literature review identified five areas most studied in current literature, including Tele-care, Electronic Health Records (EHRs), and assistive technologies noting that HIT had positively impacted clinical health outcomes. Yet, there was much less data on patient health outcomes. Based on their findings, they made recommendations to managers and stakeholders regarding how they might integrate these technologies into their organizations [237]. Vargemidis and colleagues investigated older adults' inclusion and participation in wearable physical activity tracking systems research. They found that overall older adults' participation and inclusion in this research is limited [236]. Others have also explored trends in research focused on older adults and sensing in fall detection [219], the health benefits of video games [100], technology for mental health [103], and chronic disease management [217]. These studies have identified how various technologies are being used to support older adults' health and wellness, the specific kinds of technologies that are being represented, and their overall effectiveness for addressing challenges encountered by older adults to highlight shortcomings and areas for future exploration. Among these are a need to improve research design and evaluation measures, examine cost-benefit tradeoffs, and improve use for clinical assessment.

Other efforts have examined trends related to the adoption and acceptance of certain technologies by older adults to support health and wellness [179, 184, 260], or older adults' beliefs and experiences when using technologies for health [110, 131, 203]. For instance, in 2016, Lui et al. investigated older adults' readiness for adopting smart home technologies placing particular emphasis on older adults with complex needs [155]. This systematic literature review argues that readiness for smart home technologies among older adults and older adults with complex needs was low. Therefore, there was no evidence of the usefulness of these technologies in other areas (e.g., fall detection) apart from health monitoring. Research by Or and colleagues examined variables used to investigate the acceptance of consumer health information technologies among

older adults, finding that few studies examined social, task, organizational, and factors of acceptance [179]. Sakaguchi et al. examined patient portal use among older adults to identify what has been studied and to call for additional efforts to address barriers in the design [203]. These studies tend to focus on understanding why older adults do and do not uptake health technologies designed for them to determine areas where design might be improved.

Efforts to understand trends in the use of health technologies among older adults are also wide-ranging regarding the type of technologies examined. Several reviews of literature have examined the use of video games [100], sensing and monitoring technologies [110, 155, 219], assistive technology [131, 219], patient portals [203] and disease management [103, 217] by older adults. However, others have examined trends in using certain technology to support older adults with health and aging more broadly [131, 237]. Khosaravi et al. looked broadly at how technology being labeled as assistive technology is being used to support older adults and the effectiveness of those technologies among older adult populations across those studies [131].

Therefore, systematic reviews have become an important and effective approach for understanding the state of health technology application and use among older adults. The studies mentioned earlier also identify gaps in literature and areas where more work is needed. This type of understanding will become ever more critical as healthcare technologies continue to become more pervasive and ubiquitous in our society. Our paper builds on this existing literature by examining older adults' involvement in health-related pervasive and ubiquitous computing research to identify trends about who is involved in these studies. We examine the dimensions and demographics used to define the older adult population, how they are engaged, and how this consideration may shift the narrative of aging research in HCI.

2.2 Considering Identity to Inform Narratives of Aging in HCI

Most of the research in the area of health and well-being for older adults has focused primarily on categorizing this population by age-related changes and declines – which often pushes a negative, deficit-based perspective of what it means to get older [244]. This also relegates our knowledge of older adults as users of technology as those who experience consistent challenges. The challenge of this current framing is what social scientists argue is a result of a '*Latourian divide*' in which natural scientists, engineers, and designers who engage in scholarship related to the research and development of technology sit on one side of the understanding of the aging-technology dichotomy, and social scientists sit on another and each side is not necessarily engaged with the other [185, 186]. In their book *Designing for Older Adults*, Fisk and coauthors review age-related changes in ability that should be considered when designing for the aging population, but emphasize that all older adults do not experience decline and that not all capabilities decline with age [82]. Some HCI researchers have also made aware the harmfulness of these negative perspectives, have criticized the simplistic reduction of the aging process, and have pushed to shift from these negative myths to more positive narratives that characterize aging as a natural process that is highly individualized

[51, 60, 67, 71, 96, 142, 190, 244, 259]. This work has focused on negative discourses of aging in HCI noting the harms of limiting alternate perspectives of older adults as a non-homogenous group with different life courses, lived experiences, and needs [200].

The call for shifting to more positive narratives in HCI literature has also led researchers to examine situational and intersectional factors when examining technology design with older adult populations, carving a path for constructs such as identity to be formally examined among factors of adoption and acceptance. For example, Righi and colleagues introduce the idea of situated communities as a construct for approaching research and the intersection of HCI and aging [200]. Their argument rests on the idea of older adults, not as a homogeneous group, but as individuals with unique needs and identities. Therefore, technologies should consider the dynamic needs and interests of older adults within a community. A recent workshop among older adult researchers also calls for a deeper examination of the harms of merging concepts of aging and disability as being synonymous [136]. Because traditionally, research on HCI and aging has focused on the decline, the topic of accessibility or supporting independence is often discussed in relation to aging, in many instances as one and the same. For example, Vine notes that at one point, much of the research published on aging-related topics in SIGCHI venues was published at ASSETS, the ACM's premier conference on aging and accessibility [244]. However, Knowles and Hanson note that understanding older adults primarily through the lens of accessibility can be harmful to the design process limiting the types of design opportunities that are explored and prioritized [136]. Other researchers have shared similar perspectives, highlighting the potential harms of a deficit-based lens and proposing alternative guidelines and approaches for engaging older adults in the design of health-related technologies [153, 211].

Within HCI and healthcare research, some researchers suggest that the emphasis on negative aging narratives can further limit the design of technologies that can actively support older adults in managing their health. Yuan and colleagues, in their investigation of social networking and co-production activities for health, suggest that negative or *pathogenetic* views of aging can minimize the natural process of aging and individual abilities to overcome challenges [259]. Therefore, in their work, they purposefully used a positive or *salutogenetic* approach to reveal designs that help older adults take a more active role in their health. D'Haeseleer and colleagues argue that deficit-based perspectives of aging can also lead to pitfalls in technology design where narratives shift to enforcing technology use even as it adds barriers, a potentially false sense of urgency on technology for healthy aging, and a tendency to design for home bound opposed to active, engaged individuals [65]. Yet, despite this forward movement, however, certain aging narratives are still limited in HCI. Harrington suggests that while HCI has become more intentional about incorporating different perspectives in design research, some individuals that sit at the "forgotten margins" are still underrepresented among our work [105]. Harrington defines individuals at the forgotten margins as those that are "*marginalized across multiple identities*" and are often neglected in technology design [105]. Therefore, to further shift and diversify the narratives present in aging research in HCI, there is a need to further understand the characteristics of older adults that are included in HCI design research and the methods and approaches

used to include them to better understand the narratives present in the design of health-related ubiquitous and pervasive technologies.

2.3 Exploring Identity Dimensions in HCI

There has been recent attention on understanding trends in identity dimensions that are present in research across the field of HCI [113, 138, 154, 206, 252]. In a review conducted by Schlesinger et al. researchers found that the most common identity dimension considered across a corpus of HCI research studies was gender [206]. Using a comprehensive content analysis of HCI literature published in 2006, 2011, and 2016, Himmelsbach et al. [113] find that most commonly, HCI research reports on dimensions of gender, age, and education, with dimensions such as ethnicity and culture, race, and income each reported in less than 20% of papers. Although Schlesinger identifies that 64.29% of HCI papers published between 1982 and 2016 report on examining age as a dimension of identity, we do not have an analysis of the intersections of other dimensions that overlay age. Schlesinger reports that race and class dimensions are only reported in 3.57% of papers published between this time, signifying a need for a larger shift in our research culture. Additionally, a more recent exploration of how HCI research is skewed towards more Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations found that only a small sample of literature produced from Western countries focused on individuals from lower-income geographic areas or individuals with lower education despite the ratio of non-educated citizens of that geographic area [154]. However, most HCI research studies engage mostly with ‘*primary dimensions*’—that is, dimensions that help to “shape our basic self-image and our worldview”. Loden and Rosener [156] state the primary dimensions as: *age, ethnicity, gender, physical abilities/qualities, race, and sexual orientation*. Loden and Rosener’s model goes on to define ‘*secondary dimensions*’ such as *educational background, income, marital status, work experience, religion and geographic location* as those characteristics acquired later in life. Although these secondary dimensions have less influence they still contribute to our experiences. Thus there is value in examining identity dimensions across both the primary and secondary levels.

Looking across HCI research, we see that an understanding of values and backgrounds have been studied as important constructs and predictors of the way people experience technology [28, 29, 68, 106, 108, 251]. More specifically, several scholars across health-related HCI research have noted the importance of identifying health needs and use cases for health technologies and tools among marginalized and vulnerable populations [204, 218], as well as the implications of these technologies neglecting certain subgroups [238–240]. Both Parker and Harper [96] and Harrington and Piper [108] assert the need to integrate the culture and community values of older adult groups as a part of the design approach. Yet, in the few instances where race or ethnicity is examined among older adult samples, it is most commonly to examine differences in racial attitudes to a particular intervention, presumably enforcing a notion of “othering”, or comparing ethnic minority older adults to white older adults as the standard. Rector et al. [198] establish that designing health-related technology by examining values systems of those intended to engage with the system helps to identify when and where technology should be inserted and what may be deemed

acceptable among certain populations. This indicates a need to better understand identity as a foundation for technology design research.

A more recent trend among HCI aging research is identifying the homogeneity in the older adults studied as a limitation of a research study and acknowledging a need for a more diverse representation among their sample [94, 190]. While this trend speaks to champion for a more diverse corpus of health-related HCI research, this diversity commonly refers to ability or technology background. This does not serve to shift the way these technologies are conceptualized and developed for marginalized and underserved populations. This representation matters not only to the stratification of our research samples but also for the consideration of who technologies are built for and whose values are considered in the design process.

Given the growing concern about who has access to health technologies, there are still gaps in understanding how identity dimensions are focused in HCI aging research, limiting our understanding of who and how different subgroups of older adults are included. Therefore, in this review, we examine how these aspects of HCI research are being considered and reported and at what point in the design process marginalized older adults are included in technology research and development.

3 METHOD

To gain a comprehensive overview of the trends in health-related HCI research studies that have focused on older adults, we conducted a systematic literature review [132] of full research papers published in two major computing databases. Our overall goal was to identify, evaluate, and interpret the ways health-related HCI research and the associated design efforts engaged various subgroups of the older adult population and in what ways researchers considered the various constructs that speak to their health technology experiences. We were particularly interested in research studies that focused on health-related technologies that would be used by older adults and not those used by medical personnel or healthcare professionals in a hospital setting to monitor, diagnose, or treat older adults. Systematic reviews as a research method have been proven as a prominent practice in gathering evidence on topics in the medical sciences [132, 157]; thus, there is promise for this approach to create an accurate depiction of the literature corpus in the computing disciplines. This review focused on the ACM Digital Library and IEEE Xplore Databases as leading repositories that publish research on technologies from human-computer interaction and design perspective.

Additionally, ACM has been used in conducting systematic reviews discussing various areas of computing research [70, 76, 206, 218]. Articles that appear in these venues have a high impact on the future of technology research and development both within and outside academic universities, thus were appropriate for our review. While we acknowledge that other health-related publication databases might also feature papers focused on health technologies, those databases do not emphasize HCI or design practice, so we did not include them in our review. Steps to perform this systematic review were identified by medical and health sciences librarians affiliated with the first author’s institution and adapted from [225].

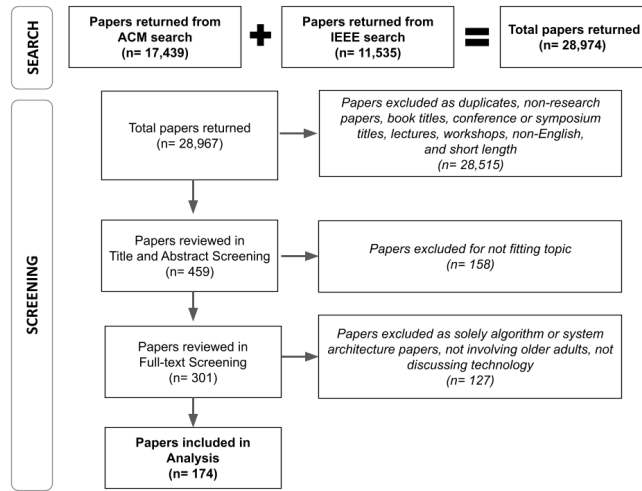


Figure 1: Search and screening procedure.

3.1 Search Process

The two first authors identified search terms of interest to aggregate a corpus of literature reflecting how the HCI community discusses health computing technologies for older adults. These search terms were combinations of the following three areas: **1 - domain** (e.g. ‘health’, ‘well-being’, ‘wellness’), **2 - population terms** (e.g. ‘older adults’, ‘senior citizens’, ‘elderly’), and **3 - intervention terms** (e.g. ‘device’, ‘system’, ‘technology’, ‘wearable’) to catch all possible papers that may be relevant. The final set of papers were pulled and filtered in February of 2021, which included research articles published between the span of January 2008 to December 2020. Figure 1 below shows the search and screening procedure¹.

We focused our systematic review on the last 12 years of aging HCI and health technology research to examine recent trends of research studies conducted among older adults. In particular, since one of the focuses and main contributions of this research is trends in reporting identity variables in older adults and health technology literature, we wanted to narrow our examination of reporting trends to the decade leading up to and the years that followed the publication of seminal papers examining intersectionality, identity dimensions, and diversity of users in broader HCI literature published in 2017 and 2019 [113, 206].

3.1.1 Title and Abstract Screening. Following the initial search, two researchers first reviewed articles and removed duplicates. Additionally, researchers screened papers to eliminate any papers determined by the page length not to be full research articles (short papers, works in progress). Full papers were defined as articles that included at least six pages of text, not including references. Researchers also excluded other papers that did not meet other exclusion criteria. Other inclusion criteria required articles to (1) be published between January 2008- December 2020, (2) be published in English, (3) focus on the design of technology primarily for use by the older adult population (opposed to technologies used by others to monitor or support older adults), and (4) discuss a

technology, device, or system to be used by older adults or a study to design such an intervention. Four researchers conducted a title and abstract screening to remove any papers that were not related to the purpose of this review. The researchers also removed papers that they determined from the abstract did not meet any inclusion criteria (discussing another population other than older adults, designing and testing an algorithm, sensor, or system only, etc.). Abstracts that contained no mention of aging, older adults, elderly, or health or wellness were also removed. If researchers suspected that a paper should be removed but still had questions or were unsure after reading the abstract, it was left for full paper screening.

We excluded papers that were (1) a book chapter, Master’s thesis, or Ph.D. dissertation, (2) non-research articles (panels, lectures, keynotes, doctoral consortium submissions, or conference abstracts), (3) works-in-progress, poster abstracts, demos, literature or systematic reviews (4) did not focus on health or wellness, (5) discussed only algorithm, sensor, or system architecture design with no mention of device or technology, (6) discussed usability heuristics from the perspective of experts, (7) discussed a framework or method with no discussion of technology, or a device (7) did not include older adults in the sample or (8) discussed the design or evaluation of patient-centered healthcare equipment or systems to be used by healthcare professionals or other caregivers.

3.2 Data Extraction

3.2.1 Full Paper Review. We reviewed titles and abstracts of 308 papers to identify the final 174 for full paper review. The research team identified a sample of papers from the larger literature corpus to establish reliability in full paper screenings. Three researchers performed six rounds of coding with 2-3 papers each round to establish reliability. During each round we made sure that all researchers understood the codes for data extraction and would extract the same data for each code. Our final reliability was ($r=93\%$). Once researchers established reliability, three researchers split the remaining papers to complete full paper screenings. One researcher fully reviewed each paper. Coders first confirmed that the paper met inclusion criteria. If papers met the inclusion criteria, the coder entered details about the paper and study(ies) into a shared spreadsheet to extract and summarize the contents of articles. If a paper was flagged for removal by the original reviewer, it was also reviewed independently and verified by another reviewer before being removed. If a reviewer questioned whether a paper met inclusion criteria, the paper was marked as “discuss,” and a final agreement of whether to include the paper in the final corpus was made among the three reviewers. Thus we ended up with 174 papers that were coded for data analysis².

3.2.2 Full Paper Coding. We developed a coding survey that included the finalized set of codes to deductively extract key data from papers and establish coding reliability. We used coding as a method because of its ability as an analytical process that identifies concepts and dimensions in data [48, 49, 254]. Researchers coded papers using a survey questionnaire which supported a hybrid coding approach following [70]. Survey questions were closed-ended and included such things as publication year, research country,

¹Search strings used for each database included as an Appendix

²a full list of papers are included as an Appendix

Type of Research Study	Exploratory Studies (n=32)	Generative Studies (n=9)	Evaluative Studies (n=109)	Mixed Studies (n=38)	Total (N=188)
Characteristics of Study Sample					
Older Adult Sample Only	75% (n=24)	66.6% (n=6)	77.9% (n=85)	78.9% (n=30)	77.1%(N=145)
Comparative w/Younger Adults	50% (n=5)	11.1% (n=1)	11.9% (n=13)	10.5% (n=4)	12.2%(N=23)
Comparative w/Kids (>18yo)	3.1% (n=1)	0% (n=0)	0.91% (n=1)	0% (n=0)	1%(N=2)
Average OA Sample Size	M=31	M=21	M=23	M=20	
Geographic Environment					
Rural Setting	3.1% (n=1)	22.2% (n=2)	0% (n=0)	0% (n=0)	1.6%(N=3)
Urban Setting	6.3% (n=2)	22.2% (n=2)	2.8% (n=3)	7.9% (n=3)	5.3%(N=10)
Mixed Setting	12.5% (n=4)	11.1% (n=1)	1.8% (n=2)	2.6% (n=1)	4.3%(N=8)
Suburban Setting	3.1% (n=1)	0% (n=0)	0% (n=0)	0% (n=0)	0.53%(N=1)
Study Setting					
Field Studies	43.8% (n=14)	55.5% (n=5)	54.1% (n=59)	36.8% (n=14)	48.9%(N=92)
Research Labs	9.4% (n=3)	0% (n=0)	17.4% (n=19)	15.7% (n=6)	14.8%(N=28)
Mixed-Site	6.3% (n=2)	11.1% (n=1)	4.5% (n=5)	5.2% (n=2)	5.3%(N=10)
Remote Study	12.5% (n=4)	0% (n=0)	0% (n=0)	2.6% (n=1)	2.6%(N=5)

Table 1: Types of Research Studies (Calculated out of 188 studies)

type of research, research setting, description of older adult sample, impairments or health conditions discussed, description of the technology designed, and description of testing. As one of the primary objectives of this review was to identify the constructs of identity included in this body of research, demographic codes were also extracted, including education, race, age, income, employment status, gender, technology proficiency, and literacy. Following guidance from medical and health science librarians at the first author's institution, coding categories were validated against related systematic reviews, and content analyses [113]. The coding survey was collaboratively refined with all authors to ensure appropriate verbiage and understanding³.

Closed-ended responses were descriptively analyzed for frequency and trends to understand the types of papers in our corpus. We used a two-phase approach to analyze open-ended responses. An initial round of coding was done using open coding, followed by a more focused round of coding to categorize emerging themes in the types of technologies reported, study outcomes, and any major findings. For example, in addition to the pre-defined categories used to code 'Study Measures' (section 4.1.2), coders entered open-ended descriptions of measures that did not fit a pre-defined category. After completing the initial coding, we used affinity diagramming to assign each of the newly identified study measures to one of the existing eight categories or combine them into new categories. Open-ended descriptions led to 46 additional study measures.

4 RESULTS

In this section, we address our research questions by discussing our results in four sections. We discuss descriptive statistics on research approach and environment, as well as categories of technological devices. To answer our research question of how HCI researchers consider and report on older adult samples and the diversity across these samples, we discuss the identity dimensions and categorical levels of these dimensions. We then report on the trends across these papers such publication rate, venues of publication, and common research methods and study design. Across the 174 papers reviewed

and coded, 12 papers discussed two or more studies with three studies being the highest number of studies reported in a single paper (n=2) [169, 172]; thus our results discuss 188 studies. We did not encounter any instances where two papers explicitly described the same study.

4.1 Research Approach, Methods, and Environment

4.1.1 Types of Research. Of the 174 papers reviewed as a part of our systematic review, 170 reported study methods. Three papers discussed design and no evaluation, and one comparative study did not discuss methods. The three papers that discussed only design presented a technology design concept proposed by the authors that was based on prior published studies. However, the papers only described the design concept and did not describe the use of any methods to evaluate or gain feedback on the design. Of the 170 papers that reported methods, 59 reported qualitative methods only, 48 reported quantitative methods only, and 63 reported quantitative and qualitative mixed methods. Nearly half, or eighty-three papers (49%) reported only one method, 16 papers (9%) reported using more than one qualitative method, eight studies (5%) used more than one quantitative method, and 63 papers (37%) used mixed-methods (both qualitative and quantitative methods) to evaluate the technology. To further investigate the type of research included in the papers, we also coded papers as being either 'Exploratory', 'Generative', 'Evaluative', a Survey, or a combination of these and defined them as follows:

- **Exploratory:** any early-stage research that has the purpose of generating requirements for design, understanding a technological, design, or societal problem, or understanding a group of users. Examples include requirements generation, needs assessments, early-stage studies to understand a problem, early-stage studies to understand a group of users.
- **Generative:** any research that has the purpose of generating one or more concepts, design, or designing/developing a prototype or interface.

³Our coding tree is included as an appendix

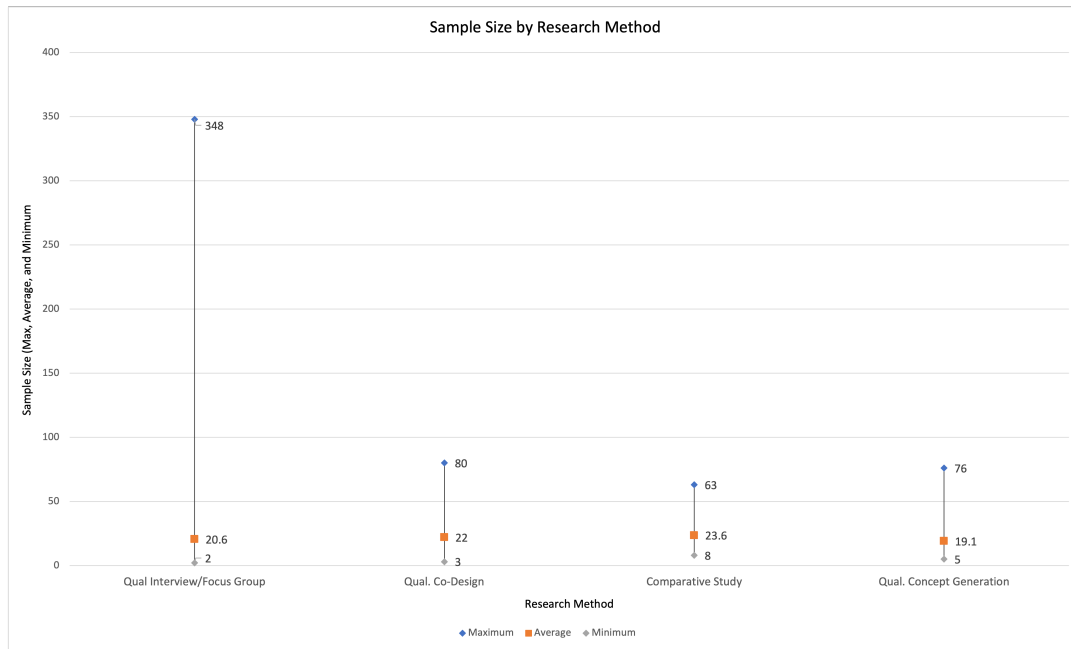


Figure 2: Average Sample Size by Research Method

- *Evaluative*: any research that has the purpose of testing, evaluating, piloting, or demoing a prototype or interface, including comparative studies.
- *Survey*: any paper that discusses collected data on the types of technologies used by a group based on age or disability.

Thirty-two studies were exploratory studies, nine were generative, 109 were evaluative, and thirty-eight studies were a combination of two or more research types. For example, [24] was coded as both “*exploratory*” and “*evaluative*”. Additionally, 30 studies compared older adults to other subgroups, with 23 papers comparing older adults to younger adults (aged 18-54) and two papers comparing older adults to youth or adolescents (aged 17 and under). Five additional papers compared older adults to another group, but the authors did not specify the age range of the comparison group in the papers (see Table 1).

Of the 188 studies, 58% (n=109) conducted qualitative interview or focus groups. Ninety studies were reported as quantitative experiments, and thirty-nine as quantitative surveys. Methods which directly engaged older adults in the design or conceptualization of technologies were reported as qualitative co-design/participatory design (n=29) and concept generation (n=16) done primarily by researchers.

Overall, study sample size comprising all participants, including non-older adults, ranged from 2 to 720 study participants ($M=31$). The sample size for only older adult participants ranged from 1 to 720 study participants ($M=27$). Participant sample size among interviews and focus groups varied the most across qualitative studies, with a minimum of 2 participants and a maximum of 348 (see Figure 2 for the average participant sample size across qualitative research methods). On average, exploratory studies had a sample

size of 31 older adults, generative studies reported an average sample size of 21 older adults, evaluative studies had an average sample size of 23 older adults, and mixed studies had an average sample size of 20 older adults. Table 1 has a breakdown of research study type, sample size, and location. Additionally, several studies (n=25) focused on aging as an inter-generational relationship dynamic between older adults and grandchildren (e.g., [84]), older adults and their adult children (e.g., [150, 182]), or as non-familial relationships with children (e.g., [125]). These studies were primarily conducted in the United States (n=7) and Canada (n=2) countries; however, other countries also looked at differences in experiences between younger and older adults.

4.1.2 Study Measures. We coded papers for the following study measures: *Usability*, *Attitudes*, *Technology Acceptance*, *Privacy*, *Usage*, *Familiarity/Experience*, *Abilities Testing* and *Falls Accuracy*. *Usability* examined measures related to how well the technology works (e.g. measures of efficiency, satisfaction, ease-of-use, effectiveness, or learnability). *Attitudes* examined self-reported beliefs such as perceptions or willingness to use. *Technology Acceptance* examined technology acceptance, perceived use, perceived usefulness, or TAM. *Privacy* examined privacy, trust, or credibility concerns. *Usage* examined frequency or duration of use. *Familiarity/Experience* examined participants’ familiarity or experiences with technology such as proficiency testing by a survey. *Abilities Testing* encompassed speed or task completion, as well as studies that discussed tests used to assess cognition and memory. *Falls Accuracy* examined fall detection accuracy with older adult participants. We acknowledge that usability and abilities testing often have overlap. We differentiated usability from abilities testing based on how the evaluation was framed by the authors (i.e., testing the

Study Measure	# of Papers	Papers
'Attitudes'	125	[1–5, 7–12, 15, 17, 23–26, 30, 33–36, 39–42, 45, 50, 53–57, 59, 61, 62, 65, 66, 69, 72–75, 78–81, 87–93, 95, 97–99, 101, 109, 114, 115, 117–119, 122, 124, 126, 129, 130, 133–135, 145, 147–150, 152, 159, 162, 163, 167–169, 171, 172, 175, 177, 178, 187–189, 191, 193, 201, 202, 213–216, 220, 223, 226–229, 231, 231, 233–235, 241, 245, 248, 250, 253, 255, 257, 262, 263, 266]
'Usability'	93	[1, 2, 4, 5, 9–11, 15, 17, 21, 26, 30, 33, 34, 37, 39, 41–44, 50, 53, 54, 56, 59, 61–63, 66, 69, 73, 75, 78, 81, 85, 87–93, 95, 97–99, 115, 119, 120, 122, 124, 129, 130, 133–135, 137, 140, 149–151, 159, 167–169, 171, 172, 175, 178, 187, 188, 191–193, 201, 205, 208, 214, 215, 222, 231, 232, 235, 241, 247, 248, 253, 255, 256, 262–264, 266]
'Usage'	32	[17, 26, 30, 39, 40, 53, 56, 59, 62, 66, 69, 84, 87, 95, 97, 115, 118, 122, 124, 125, 133, 150, 167, 175, 189, 191, 201, 202, 208, 214, 215, 223]
'Abilities Testing'	30	[1, 4, 14, 41, 42, 44, 50, 54, 78, 80, 81, 88, 95, 119, 123, 143, 149, 152, 159, 170, 175, 194, 202, 222, 231, 233, 246, 247, 250]
'Technology Acceptance'	20	[2, 10, 11, 17, 30, 33, 45, 50, 63, 79, 97, 126, 133, 151, 201, 215, 231, 248, 253, 266]
'Familiarity/Experience'	8	[15, 56, 97, 144, 175, 193, 233, 263]
'Fall Detection'	5	[86, 159, 162, 232, 253]
'Privacy/Trust'	3	[85, 212, 214]

Table 2: Breakdown of Study Measures and Associated Papers

interface or testing the abilities of the user). For example, some studies examined whether an interface allowed users to complete tasks efficiently or to a users satisfaction whereas other studies tested user ability to complete a task efficiently using the interface.

Of the 188 studies conducted across papers, 125 examined participants' attitudes toward technology (see Table 2). Ninety-three examined usability, 32 examined usage, and 30 examined users' abilities or ability to use technology. Twenty studies examined technology acceptance, eight measured familiarity or prior experience with technology, five measured fall detection accuracy with older adult participants, and three examined privacy and trust. Six studies did not explicitly state the measures collected. Fifty-seven studies included only one measure, and 111 included multiple measures.

In addition to pre-defined categories of study measures, open-ended descriptions led to 46 additional measures. Eighteen of the 46 codes were related to the original eight categories. Studies that related to pre-defined codes were re-coded using the appropriate higher-level category as needed. Twenty-eight codes were merged and this process resulted in an additional eight categories: *'Recognition accuracy/Accuracy'*, *'Features of rehabilitation practice'*, *'Desired features/Functionality'*, *'Affective/Emotional behaviors'*, *'Social interactions/Behaviors'*, *'Health metrics'*, *'Health/Behavior change'*, and *'Visual imagery'*. *Recognition accuracy/accuracy* included measures of accuracy not related to falls accuracy (n=5). Studies examining *affective/emotional behaviors* focused on collecting data on mood, EEG, mental state, or similar measures (n=6). *Social interactions/behaviors* included measures related to social presence, social support, loneliness, or similar (n=14). *Health metrics* such as physical activity, physical metrics or the use of surveys to determine health state or risks (e.g., risk of depression, COPD risks) were collected in nine studies (n=9). *Health/behavior change* studies examined adherence to a health regimen such as medication routine (n=6). Three studies collected data about desired features/functionality. One study each collected data about rehabilitation practice and a visual imagery measure.

4.1.3 Study Location and Duration. Studies spanned 34 different countries. Most research studies (n=53) were conducted in the United States, followed by Canada (n=14), and then Germany and the United Kingdom (n=12 each). Nine were conducted in The Netherlands, eight were conducted in Portugal, seven in Australia, and six were conducted in both Japan and Singapore.

Of the 188 studies examined, 165 did not explicitly state whether they were conducted in a rural, urban, mixed, or suburban environment. Ten studies indicated that recruitment occurred in an urban environment, nine indicated a mixed environment (e.g., rural and urban, rural and suburban), three indicated a focus on a rural environment, and one explicitly mentioned a focus on suburban environments (see Figure 3).

To better understand how previous literature evaluated technology in different living environments, we coded for the research site to capture which studies were conducted in a research lab, at a field site (e.g. participant's home, community center, health clinic), or a mixed site between lab and field. Forty-three studies did not explicitly state study location. Of the studies that reported the study location (n=142), 30 were done in research labs, 95 were field studies, 6 were done remotely, and 11 reported having a mixed-site approach.

Twenty-one of the 188 studies did not discuss testing or evaluation. Of the studies that conducted testing or evaluation of a device or prototype, 58 studies did an evaluation for less than a month, 30 evaluated technology for between two months and five months, seven evaluated a technology between 6 months to a year, and seven studies evaluated the technology for more than a year. Sixty-five studies discussed testing but did not state the study duration.

Field studies were most common among exploratory (n=14), generative (n=5), and evaluative studies (n=59). Lab studies were the next most common among evaluative studies (n=19), while remote studies were more common among exploratory studies (n=4). Seven exploratory studies did not specify the type of site used, sixteen evaluative studies did not specify, and two generative studies did not explicitly specify. Fourteen mixed-site studies were conducted in the field, six were conducted in the lab, and 11 studies did not specify (see Figure 3).

4.2 Categories of Technologies and Devices

4.2.1 Descriptions of Technology. Descriptions of technology were pulled directly from the text. Robots were among the most mentioned technology descriptions and were directly referenced 32 times within the literature. These included descriptions such as "assistive robot," "social robot," and "mobile service robot". Games were the next most commonly mentioned, described in 29 studies, followed by apps which were described in 21 studies. Common

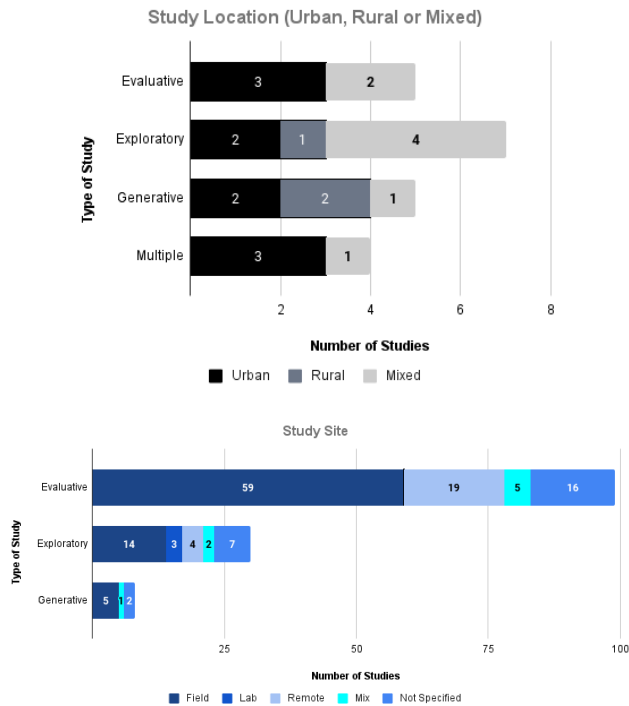


Figure 3: Left: Study sites split by type of study. Right: Study locations split by type of study.

descriptors of technologies across the research studies we reviewed included “assistive” (n=18 studies), and “social” (n=16 studies).

Categories of technology were reported in our review as explicitly defined by the authors (see Figure 4). Out of the 188 studies we reviewed, 181 reported using specific technologies in their studies. Seven papers did not report the use of a category of technology category or device, and instead evaluated the needs of target populations among older adults when considering prospective technologies or demographic subsets. Six of these were exploratory, and one was generative. We differentiate technology versus the device described as technology being any product or system, and device being the medium used for deployment. While most of the studies reported a single technology category, 72 of the studies included more than one. In keeping with the device types, the most often reported technology categories were those classified as mobile devices or applications (23.40%, n=44). Closely following were games, reported in 36 studies, personal robot assistants in 24 studies, and IoT devices in 19 studies. Seventeen studies were classified outside of our major categories. Researchers also added additional in-text classifications for technologies that fell outside of these categories, such as social robots (n=1), web-based health tutorials (n=1), and ambient displays (n= 1).

4.2.2 Type of Devices. To understand the kinds of devices being created with and for older adults, we further evaluated the devices mentioned in each study. Of the 188 studies we reviewed, 181 discussed devices, and 60 studies mentioned more than one device. The

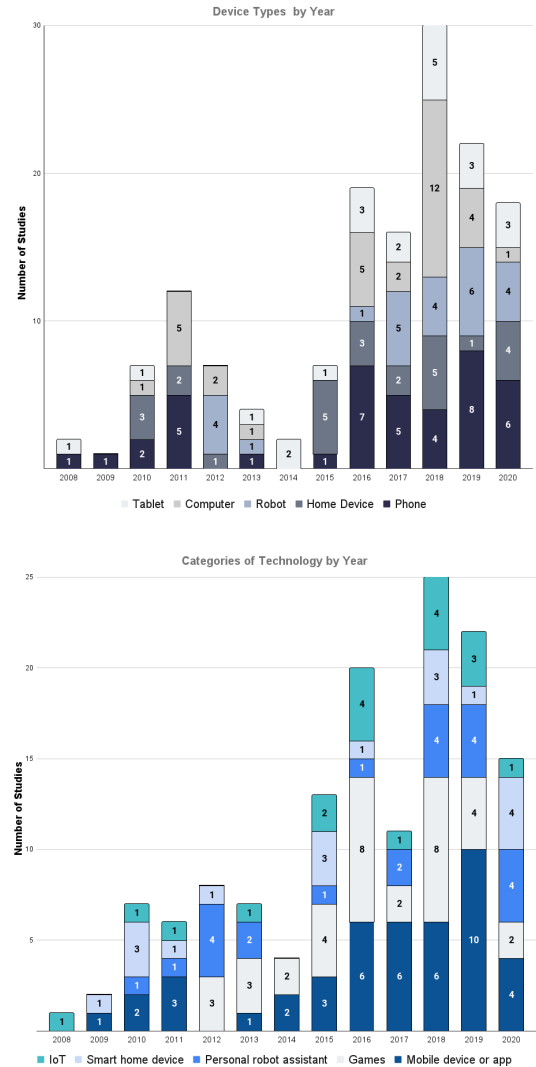


Figure 4: Left: Top 5 devices out of 181 studies, split by year. Right: Top 5 technology categories out of 181 studies, split by year.

majority of studies used a single device. Studies that mentioned devices were coded by category, type, and in-text descriptions. Types of devices were reported as explicitly stated in the text of the papers. The most prevalent devices were mobile phones (n=41), followed by computers (n=33) and home devices (n=26). (See Figure 4).

4.3 Population Identity Dimensions Reported

Based on Loden and Rosener’s work [156], as well as the work of Himmelsbach et al. [113], we observed and coded papers for both ‘Primary Dimensions’ (Age, Ethnicity & Culture, Gender & Sex, Mental Abilities, Physical Abilities, Race, and Sexual Orientation) and ‘Secondary Dimensions’ (‘Appearance & Body’, ‘Class’, ‘Education’, ‘Geographic Location’, ‘Language & Fluency’, ‘Migration Biographies’, ‘Parental Status’, ‘Relationship Status’, and ‘Religion’).

Type of Research Study	Exploratory Studies (n=32)	Generative Studies (n=9)	Evaluative Studies (n=110)	Mixed Studies (n=37)	% out of all 188 studies (Total N)
Primary Dimensions					
Age	93.7% (n=30)	89% (n=8)	94% (n=104)	86.5% (n=32)	92.5% (N=174)
Gender	75% (n=24)	55.5% (n=5)	69.1% (n=76)	81.1% (n=30)	71.8% (N=135)
Disability/Impairments	18.7% (n=6)	11.1% (n=1)	38.2% (n=42)	21.6% (n=8)	30.3% (N=57)
Race/Ethnicity	15.6% (n=5)	0% (n=0)	9.1% (n=10)	18.4% (n=7)	11.7% (N=22)
Health Status	15.6% (n=5)	11.1% (n=1)	9.1% (n=10)	10.8% (n=4)	10.6% (N=20)
Home Support for ADLs	6.2% (n=2)	0% (n=0)	4.5% (n=5)	8.1% (n=3)	5.3% (N=10)
Fall History	3.1% (n=1)	0% (n=0)	3.6% (n=4)	2.7% (n=1)	3.2% (N=6)
Medication History	0% (n=0)	0% (n=0)	1.8% (n=2)	2.7% (n=1)	1.6% (N=3)
Secondary Dimensions					
Education	25% (n=8)	0% (n=0)	23.6% (n=26)	10.5% (n=4)	20.2% (N=38)
Environment	18.7% (n=6)	44.4% (n=4)	5.4% (n=6)	10.5% (n=4)	10.6% (N=20)
Home Living Status	12.5% (n=4)	11.1% (n=1)	10% (n=11)	7.9% (n=3)	10.1% (N=19)
Occupational Status	9.4% (n=3)	11.1% (n=1)	6.4% (n=7)	5.3% (n=2)	6.9% (N=13)
Social Support	6.2% (n=2)	11.1% (n=1)	5.4% (n=6)	7.9% (n=3)	6.4% (N=12)
Exercise/Activity Level	0% (n=0)	0% (n=0)	5.4% (n=6)	10.5% (n=4)	5.3% (N=10)
Previous Profession	9.4% (n=3)	0% (n=0)	5.4% (n=6)	0% (n=0)	4.8% (N=9)
Marital Status	9.4% (n=3)	11.1% (n=1)	1.8% (n=2)	2.6% (n=1)	3.7% (N=7)
Income	3.1% (n=1)	11.1% (n=1)	2.7% (n=3)	5.3% (n=2)	3.7% (N=7)
Appearance and Body	0% (n=0)	0% (n=0)	4.5% (n=5)	0% (n=0)	2.6% (N=5)
Parental Status	3.1% (n=1)	0% (n=0)	1.8% (n=2)	2.6% (n=1)	2.1% (N=4)
Language and Fluency	0% (n=0)	0% (n=0)	2.7% (n=3)	0% (n=0)	1.6% (N=3)
Health Provider Relationship	6.2% (n=2)	0% (n=0)	0.9% (n=1)	0% (n=0)	1.6% (N=3)
Health Literacy	0% (n=0)	0% (n=0)	0.9% (n=1)	2.6% (n=1)	1.1% (N=2)
Health Behavior	0% (n=0)	0% (n=0)	0% (n=0)	2.6% (n=1)	0.5% (N=1)
Personality	0% (n=0)	0% (n=0)	0.9% (n=1)	0% (n=0)	0.5% (N=1)
Tech Use Characteristics					
Tech Proficiency/Familiarity	34.4% (n=11)	66.7% (n=6)	26.4% (n=29)	24.3% (n=9)	29.2% (N=55)
Assistive Technology Use	6.2% (n=2)	11.1% (n=1)	6.4% (n=7)	8.1% (n=3)	6.9% (N=13)
Internet/Computer Use	3.1% (n=1)	22.2% (n=2)	7.3% (n=8)	0% (n=0)	5.8% (N=11)
Health tracking	0% (n=0)	0% (n=0)	2.7% (n=3)	0% (n=0)	1.6% (N=3)
Technology Ownership	3.1% (n=1)	0% (n=0)	1.8% (n=2)	0% (n=0)	1.6% (N=3)
Wifi Access	0% (n=0)	0% (n=0)	0.9% (n=1)	2.7% (n=1)	1.1% (N=2)
Interest in Technology	3.1% (n=1)	0% (n=0)	0% (n=0)	0% (n=0)	0.5% (N=1)

Table 3: Identity Dimensions by Types of Research Study (% out of Research Study Type).

We also captured any additional identity dimensions that may have appeared. In addition to Primary and Secondary dimensions, we also captured dimensions that described a participant sample's skill or aptitude with technology prior to testing an intervention such as technology and work experiences, recreational habits, digital literacy, and technology proficiency. Many of these were used to identify a third dimension of 'Tech Use Characteristics'. Thirty-one identity dimensions appeared across the corpus of literature we coded. Table 3 details each category of identity dimension by type of research study and then the total number of times that dimension appeared in research studies across our corpus. We discuss the descriptive statistics of each below.

4.3.1 Primary Identity Dimensions: Age and Gender Demographics. The definitions of 'older adults' often vary across different countries and cultures. In the United States, the U.S. Census considers those 65 years of age or older as older adults [242] which corresponds with retirement age. The World Health Organization begins at 60 years [180], while some places in Europe consider individuals 50-55 as older adults [230]. The lowest average age of participants in the studies we reviewed was 55 [133], and the lowest reported age range in studies of older adults started at 34 [78]. On average, most studies saw more women older adult participants than any other gender representation. However, we note that few if any papers identified gender beyond the male/female binary or even reported a category of "other" when discussing gender [158, 174].

4.3.2 Primary Identity Dimensions: Health Conditions and Impairments. Researchers have previously discussed how technology might accommodate for older adults aging into and/or with disabilities and how conceptions of age and disability are factored into the design of assistive or rehabilitative technology [164, 199, 244]. Among the studies reviewed, 57 (30.3%) collected and reported on the sample's functional or impairment status. Of those 57 studies, 37 (64.9%) identified specific forms of disabilities, and 53 (93%) reported chronic health conditions. Of the studies that reported a particular health condition, the most commonly reported condition was Alzheimer's/Dementia (n=17) followed by Depression (n=5). Figure 5 shows the frequency of health conditions reported across studies.

4.3.3 Primary Identity Dimensions: Race, Ethnicity, and Cultural Background. Across the papers that we reviewed, we noted that 'Race' and 'Ethnicity' were used interchangeably and often conflated with Nationality. This dimension was reported in 10.6% (n=22) of studies overall (see Table 3). In most instances of these studies (n=15), groups were homogeneous to one race or ethnicity, and only 4 studies explicitly engage with the race of participants as it intersects with technology requirements or usability [91, 194, 255, 256] as opposed to stating ethnicity/nationality broadly by defining recruitment location. Despite most papers using a homogeneous sample, this was not noted as a limitation. For example, a study by Jelen et al. [122] speaks to a lack of diversity in gender or educational

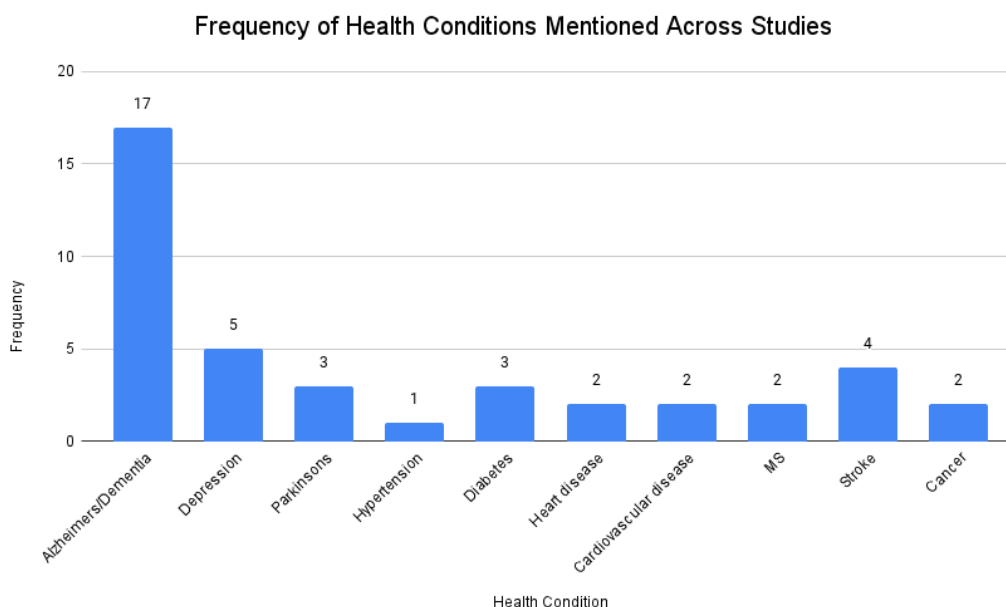


Figure 5: Frequency of Health Conditions Reported Across All Studies

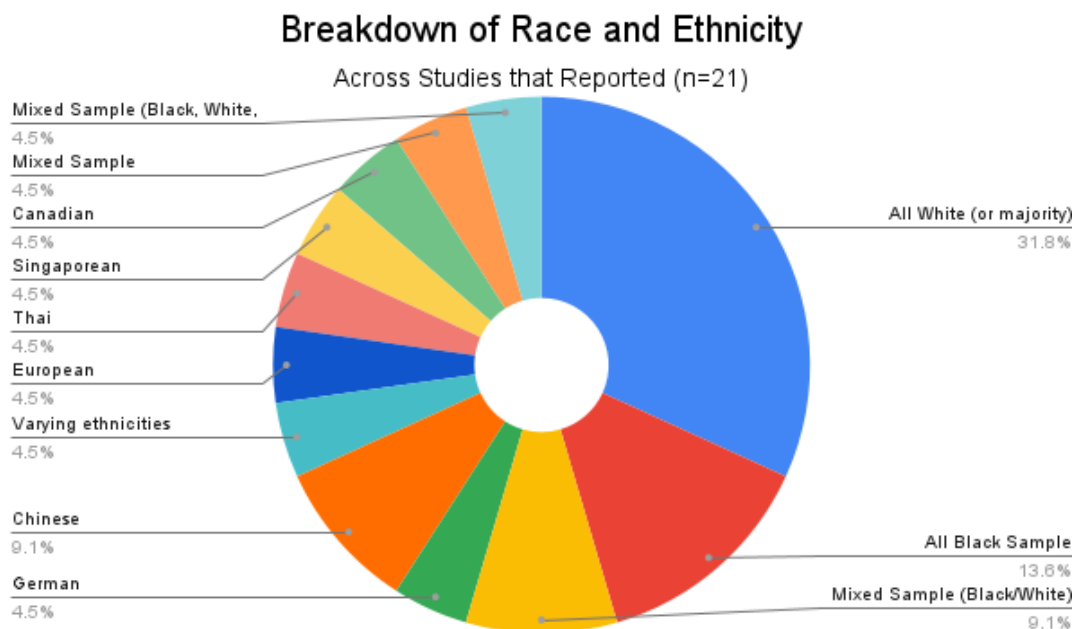


Figure 6: Breakdown of Race and Ethnicity out of Studies that Reported this Dimension (N=22)

background within their sample but do not acknowledge a lack of racial diversity. Few studies (n=2) discussed race as a determinant of study outcomes or even as a background determining factor that informs the research [117, 183].

Among studies that explicitly stated the race of older adult participants, seven report all or mostly White samples [8, 120, 122, 144, 173, 175, 194], four report a sample of all or “majority” Black older adults [56, 117, 255, 256], and four state that their sample had a

mixture of races [109, 183, 215]. Only eleven studies, 5.8% of our overall sample, explicitly reported engaging non-White older adults - [56, 91, 109, 117, 146, 183, 215, 216, 255, 256, 263]. Among studies that reported ethnicity or nationality of older adults, participants were stated to be German [245], Chinese [91], European [66], Thai [216], Canadian [84], and Singaporean [263] (see Figure 6).

4.3.4 Secondary Identity Dimensions. Among the publications and studies reviewed, we identified 15 secondary identity dimensions: 1 - Income or Socioeconomic Status; 2 - Education; 3 - Health literacy; 4 - Current occupational status; 5 - Home living status; 6 - Parental status; 7 - Marital status; 8 - Living environment; 9 - Social support; 10 - Previous profession; 11 - Personality; 12 - Fitness status; 13 - Health experience; 14 - Language and fluency; and 15 - Body and appearance (see Table 3). Of particular interest was the frequency of reporting identity dimensions which would represent diverse samples of older adults in health technology research that may directly benefit from cutting-edge technological innovations.

Of the secondary identities reported across these studies, education (n=38), living environment (n=20), home living status (n=19), occupational status (n=13), social support (n=12), and fitness level (n=10) were reported in at least 10 papers (see Table 3). The least mentioned secondary identities were 'literacy and fluency' in three studies, and 'health literacy' reported in two studies. We also noted that other identity dimensions defined by [113, 156] such as 'Religion' and 'Migration' were not mentioned in the studies analyzed.

4.3.5 Other Identity Dimensions. As a part of our analysis of identity dimensions and demographics, we observed a third level of identity that described an individual's experience with technology. We define this as a third level separate from Primary or Secondary dimensions since it does not fit in either but is still relevant and among the common dimensions captured in background and demographics data in many studies. *Tech Use Characteristics* were reported in 30% of studies (n=60). The most prevalent of these dimensions was 'Technology Proficiency and Familiarity' which was reported in 29.2% of studies (n=55) which was then followed by 'Assistive Technology Ownership' in 6.9% of studies (n=13). Other dimensions associated with *Tech Use Characteristics* that were identified in this analysis but represented in under 3% of studies included 'Wifi access', 'Interest in technology', 'Health tracking behaviors', and 'Computer ownership'. See Table 3.

4.3.6 Diverse Identity Dimensions. To better understand the trends of identity dimensions that speak to more diverse samples across this research, we isolated identity dimensions identified by Stowell et al [218], Himmelsbach et al [113], and Schlesinger et al [206]. Previous scholars have acknowledged that effective health interventions should engage the "unique experiences, attitudes, and linguistic nuances" of various cultural groups [206, 218], and thus must better identify dimensions of race, socioeconomic status, language and fluency, and education. Table 4 details studies that report on participant samples with intersections of race/ethnicity, income, education, and language/fluency. There were no studies that reported on race/ethnicity, income and language/fluency.

4.4 Trends in Research 2008-2020

Of the 174 papers reviewed, 31 were journal papers, and 142 were conference proceedings. The mostcommon conference venues

present among our final reviewed papers were *PervasiveHealth* (N=23), *ACM Conference on Human Factors in Computing Systems (CHI)* (N=18), and *PERvasive Technologies Related to Assistive Environments (PETRA)* (N=7). As shown in Figure 7, the number of studies published on health technologies for older adults saw a sharp increase from 2017 to 2018, although we note that this may reflect the number of papers that met our inclusion criteria and not necessarily the entire number of papers published in those years.

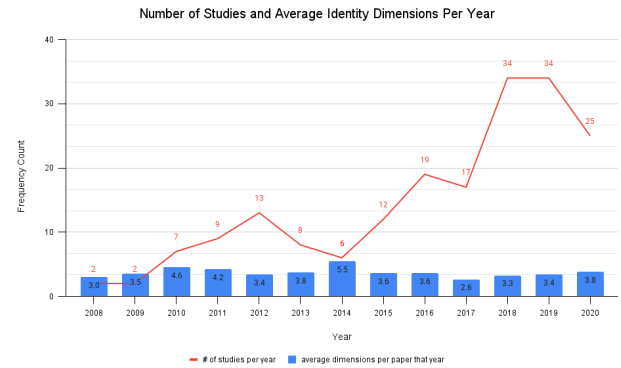


Figure 7: Number of Studies and Average Number of Identity Dimensions Per Study Each Year from 2008 through 2020.

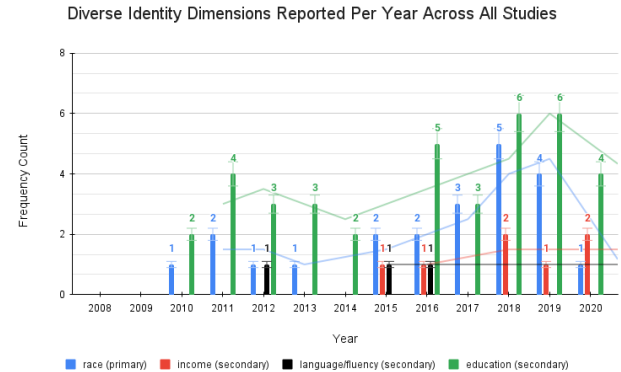


Figure 8: Total Frequency of Diverse Identity Dimensions Reported Each Year Across all Studies.

Figure 7 also shows the average number of identity dimensions per study each year. We see that there has been a slight decrease in the average number of identity dimensions per study reported to describe older adult samples across the years of this analysis, despite an increase in publications in this area. To understand the shifts in HCI research to report on more diverse samples, we again analyzed the prevalence of race/ethnicity, income, education, and language/fluency over the year. Figure 8 shows a trend of participant education being reported across more studies over the years compared to the other dimensions, followed by race. This chart also indicates more papers reporting participant income after

Intersections of Dimensions	# of Papers	Overall % out of 188 studies	Examples
Race/Ethnicity & Education	8	4.2%	[5, 122, 173, 216, 255, 256]
Income & Education	3	1.6%	[8, 197, 223]
Race/Ethnicity & Language/Fluency	3	1.6%	[56, 175, 194]
Race/Ethnicity & Income	3	1.6%	[8, 84, 117]
Race/Ethnicity & Income & Education	1	.5%	[8]
Race/Ethnicity & Income & Language/Fluency	0	-	

Table 4: Number of studies that reported race/ethnicity, income, education, and language/fluency.

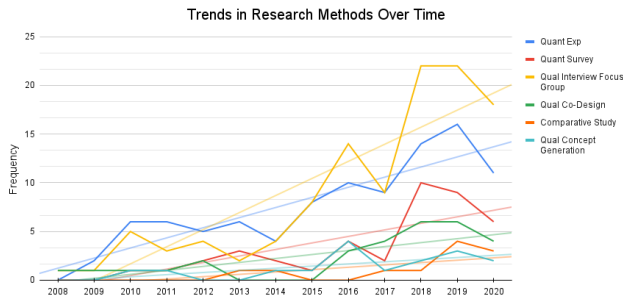


Figure 9: Trends in Research Methods Over Time by Type of Method.

2015. Overall, there has been an increase in reporting of these four dimensions across the studies we analyzed from 2008 to 2020, with an upward trajectory of dimensions reported per study following Schlesinger et al’s findings and recommendations that HCI studies observe the interaction of race, class, and gender [206].

We noticed a trend in more papers and studies focusing on interviewing older adults or evaluating technology with them, with fewer studies doing co-design or concept generation (see Figure 9). Research studies that reported older adults’ race and/or ethnicity appear to increase through 2019, while language/fluency remained low in prevalence of reporting. In terms of intervention, nearly 1/4 of the studies we reviewed focused on mobile phones or apps, with computers and games closely following.

Although we did not include them in our analysis, we noticed a trend in the number of papers that have been published in the area of health and aging that were either systematic reviews ($n=14$) (i.e. [13, 18, 258]), or papers that solely focused on sensors to detect gait or described a system architecture ($n=65$) (i.e. [6, 261, 265]). We also noted the number of papers that were returned in this search that focused on technologies for aging but did not include older adults as a sample and instead designed and tested with younger adults, college students, or other designers ($n=9$) (i.e. [47, 83]). These papers were not included in the 174 papers analyzed.

5 DISCUSSION

Our primary goal with this systematic review was to understand the ways older adults are engaged in health technology research in HCI, and what identity dimensions are considered when designing these systems. From our review we found that there has been a trend of

both exploratory and evaluative research with older adults, with most studies deploying and testing some form of mobile device/app or game. Among the identity dimensions reported across these studies, we found that research in this area has reported on primary dimensions of *age* and *gender* fairly consistently, while other primary dimensions such as *race* and *ethnicity* are under-reported, and *sexual orientation* was not reported among the studies we reviewed. It is also important to note that race and income were reported the least among evaluative studies across all types of research. Our findings also highlight a low incidence of studies that report on or consider intersections of race, socioeconomic status, and educational background, supporting similar broader reviews by [113] and [206]. Our analysis of study location and environment highlight that 88% of studies do not explicitly identify geographic environment of studies, an important insight given the prevalence of field studies reported across the corpus of our review. We also see lower number of studies that report exploring health technologies with older adults in rural environments, suggesting little improvement to the overall landscape of addressing health barriers and disparities in less densely populated areas [218].

As an emphasis on understanding who we are designing for and the identity characteristics of users has become increasingly important in the HCI community [113, 206, 218], our review serves to call to attention a major lack in aging research. While we did notice a slight increase in the reporting of certain dimensions over the years such as race and income, these dimensions are still not considered valuable in our research culture. Since such pivotal calls to action from Stowell et al. [218], Himmelsbach [113], Schlesinger [206] and others, we observe that the needle on consideration and reporting of primary identity dimensions has not moved much, and many secondary dimensions have still not made it into the norms of data collection and representation. Findings from our analysis indicates the necessity and criticality of ‘intersectional HCI’ and raise question to how well historically marginalized older adults are present and engaged in our research, and to what end are we perpetuating cycles of inequity and exclusion within HCI.

5.1 Who Participates in Health and Aging HCI Research?

Exclusion of certain groups can be harmful when developing technologies that are intended to address the needs of all users, particularly when those technologies are particularly useful interventions for historically excluded communities. We have seen how omission of certain groups can be particularly harmful in other areas of HCI

research such as facial recognition or algorithm bias [102, 207]. Because health-related technologies are designed with the goal of improving individual and community health outcomes in our society, there needs to be more attention in health-related research to explicitly consider the intersections of race, age, socioeconomic status, and educational background. Our findings suggest that in studies that examine health technologies for older adults, age and gender are reported at far higher rates than race or income. As others have identified the potential harms of reducing aging to a homogeneous experience [136], there is also potential harm in not considering the unique experience, needs, and challenges of those with different identity dimensions when designing health technologies. Considering that the methods employed in design research also play a significant role in the equity of studies [160], these considerations should also apply when examining how the methods we adopt support inclusion or exclusion of individuals with different identity dimensions. For example, recent studies on introducing older adults to customizing pervasive health technologies through crafting may promote older adults being more involved in conceptualizing and physically creating health technology [121, 122], yet much of this research has not been extended to historically marginalized older adults.

Benda and colleagues argue that beyond technology use, non-inclusive health technology design practices can lead to “intervention-generated” health inequities [22]. When this happens, health technologies end up advantaging those individuals already privileged compared to disadvantaged groups. These harms occur because the technology that was designed to reduce inequities end up widening the gap, similar to the aforementioned work on DIY crafting with older groups. Because of this, several researchers have suggested approaches for supporting more equitable design practices. For example, Montague and colleagues have suggested using culturally informed design for health technologies [165, 166]. Prior work has outlined approaches for more equitable, community-based participatory design practices [106, 108]. However, our findings indicate that health technologies research focused on older adults currently lacks inclusion or reporting of identity dimensions that would support researchers’ reflection of approaches and methods. Therefore, we argue that HCI researchers should broadly reflect on barriers to considering subgroups of older adults as a way to understand whether we might also be creating more barriers through our recruitment, methods, or research practices. Expanding the number of participants with different identity dimensions in our research could lead to more equitable technologies.

5.2 Implications for Shifting Research Practices

Duque et al. [76] argue that a prominent way to support an improvement in the design of technologies to support older adults is to consider user-centered and participatory design methods over that of traditional research methods. The authors assert that UCD/PD ensure that older adults are better integrated into the design and development of technologies that are intended to support their health. While literature in HCI supports the adoption of participatory design methods as an engagement which provides older adults with more buy-in to the design process [57, 109, 160, 161, 195], these

methods do not account for diversity in perspectives among the older population. Cozza et al. [52] state that by encouraging older adults’ direct participation in design, HCI research can potentially avoid possible stereotypes that rely on the notion that the aging population is a monolith or homogeneous group only concerned with functional ability. While this assertion is in fact valid, it positions non-traditional research methods as the sole catalyst of shifting this perception, with little critical consideration to *who* is engaged in these studies, and to what extent participation is made inclusive or accessible. Ogbonnaya-Ogburu et al. [176] argue that considering a more critical perspective in our research “offer unique insight into the world that others may not see” and “can also lead to more creative designs”.

Engaging critical theoretical frameworks such as intersectionality and Critical Race Theory may shift our research practice to better consider older adults that have been pushed to the margins of our design considerations. In its initial introduction in HCI, intersectionality was used as a theoretical framework to identify and acknowledge which categorical dimensions of identity were present and prevalent among HCI research studies, and which populations are observed [138, 206, 252]. More recent engagements with this framework argue that a more valid engagement with intersectionality is not just multiple, categorical identity dimensions, but an understanding of how these identities experience systemic oppression due to race, gender, class, sexual identity, disability status [139, 176, 196]. Scholars examining health disparities and inequities from a public health lens argue that one cannot publish on racial health inequities, associated determinants, or interventions without also understanding the impact of race and class [27]. This situates ongoing HCI research that seeks to address health challenges with technological advancements as an area that is either mitigating or amplifying the problem of health inequalities.

Thus, considering ways to shift our research practice in who is included and ways of reporting on these studies is pertinent and timely. Existing recommendations such as clear reporting of demographic characteristics and exploring more nuanced levels of identity already set a useful foundation for a better consideration of identity in health technology research more broadly [113, 158]. When considering this area of research with older adults, we contribute the following additional recommendations:

- (1) ***Eliminate “othering” by including identity dimensions of all older adult samples.*** In many of the papers we reviewed, race and socioeconomic status were reported only when studies specifically aimed to address these populations. As described by both Westerlund [249] and Law [141], our research methods work to repress that which does not fit the standard by the absence or exclusion of certain things, thus creating a sense of “othering”. Providing context of these dimensions for groups of all racial backgrounds helps researchers to frame why and how these populations may use or abandon technology, and may also serve to reduce othering in how we discuss certain populations [77].
- (2) ***Embrace the full humanity of research participants.*** Previous research asserts that by employing a deficit-based perspective of older adults, we reduce them to disabilities

and impairments, or problems to be solved. Normalizing providing more identity dimensions than just age and gender may help us to consider a more humanistic view of older adults engaged within HCI research [174]. This may also produce better discussions of how collected identity dimensions impact certain health outcomes.

- (3) **Clearly identifying the limitations of the study sample.** Clearly identifying who is included and subsequently who is not in a study can help to limit over-generalizations of findings to the broader older adult population. Examining who participates in the design process is a cornerstone of HCI research [128, 243]. Yet, we find that a deeper examination of participants' identity dimensions is often limited. We know that these dimensions can influence access to technologies [22]. Considering those older adults not included in a study and why [19, 20, 221]) can lead to better reporting of why a study does not include more diverse participants supporting reflective practice [209, 210] among the research team which can, in turn, encourage researchers to thoughtfully consider the diversity of their participants upfront when designing their study.
- (4) **Identifying the impact of chosen methods on equity.** Deciding how to engage participants (i.e., choosing a method) is often one of the first steps of study design. Selecting a particular method can often influence the feedback received and how participants engage in the design process [104]. Just as reflecting on methods can help researchers critically examine who participates in a research study, reflecting on how a chosen method impacts the inclusion of those that the research might impact can be an essential step toward more equitable research and design practices. For example, while online survey research can allow for a potentially larger research pool, it inherently excludes those without Internet access. Examining choice of method and how it affects who can and cannot share their voice can help researchers develop better strategies for including diverse populations for more equitable design.

Positioning these recommendations and others from [113, 158, 174] as foundational to doing research that directly impacts older adults better situates HCI to obtain a more nuanced understanding of people and the technology that is intended to support them.

5.3 Limitations and Future Work

This systematic review represents a focused investigation of identity dimensions used in HCI research to frame the design of pervasive and ubiquitous health technologies for older adults. Because our focus aimed at aging and technology research and how it is being framed in HCI literature, our literature corpus is limited to papers available through the IEEE and ACM databases which index the majority of computing research literature. We acknowledge that other databases may include papers that intersect HCI and design, however to scope our review to our intended questions, those papers were not included. Further, even if other databases include papers that do a better job of addressing identity variables, the fact that they are being published outside of the two major computing databases of focus may reflect biases and limitations

within the community. Similarly, we acknowledge that aging research in HCI has a long history, yet our review is limited to the last twelve years. We limited the time frame for scoping purposes and to narrow our focus to more recent trends. Additional analysis would be needed to determine if papers published outside of this time frame show similar or different trends. Our analysis is also limited to papers written in English. Although we encountered papers written in other languages, they were excluded as English was the common language among the researchers involved in the review. Future reviews would be needed to determine if papers published in languages other than English follow similar trends for reporting identity dimensions.

Our review aligns with recent calls within the HCI community to understand who is engaged and included in HCI research. Future research on identity dimensions in pervasive and ubiquitous computing research for older adults must look further than a binary classification of whether certain dimensions are collected in reporting the sample but also how these identity dimension engage and impact the computing devices and systems being designed. Critical reflection of who is engaged in the design of health technologies and why may lead to better understanding of socio-technical factors that may impact deployment of the technology in real world environments. Further, expanding who is included, may help to address health inequities by improving access to information and resources through inclusive technology design.

6 CONCLUSION

The goal of our systematic review was to understand the identity dimensions and methodological approaches reported in health-related technology design research conducted with older adults. We reviewed and analyzed 174 papers to uncover how they discuss their research practices, participants' identity dimensions, and how older adults were included and engaged in the research. Our analysis revealed that from 2008 to 2020 (the years included in our review), the number of technology design-related studies that included older adults experienced an upward trend with a slight downward trend in more recent years. While the types of research varied, the majority were evaluative focused on investigating attitudes toward technology and usability. However, in that same time frame, age and gender remained the top (> 70% of studies reporting) identity dimensions reported with few apart from technology proficiency (29%) and disability (29%) being reported in 25% of more of the papers we reviewed. Therefore, our findings suggest that many older adults with different experiences based on identities other than age or gender may be excluded out of conversations about health technology design. Given the potential impact of "intervention-generated" health inequities on access to care, treatment, and health outcomes [22] we argue that we may be exacerbating an already widening gap in access to health resources. Therefore, as designers, researchers, and practitioners tasked with designing the next generation of health technologies, we must reflect on how our research practices support inclusion and find opportunities to engage those currently not engaged. We suggest this as a first step in moving towards more equitable research for the aging population.

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